Charles Bradley

(262) 825-7432 | 24bradlc@gmail.com | linkedin.com/in/charles-bradley |Brookfield, Wisconsin

Education

California Polytechnic State University San Luis Obispo, CA
Bachelor of Science in Mechanical Engineering - Mechatronics Concentration *Expected May 2028*

- **GPA (Major):** 3.52 / 4.0
- **Relevant Coursework:** Mechanics of Materials, Engineering Dynamics, Engineering Statics, Thermodynamics, Manufacturing Processes: Material Removal and Metal Casting
- **Activities & Societies:** Professional Project Chairman, Sigma Phi Delta Engineering Fraternity

Projects

3D-Printed Fixed Wing Aircraft | *Sigma Phi Delta* Sept 2026 – Present

- Leading a fraternity-wide engineering team in the design, manufacture, and testing of a fully 3D-printed fixed-wing aircraft for the CSU 3D-Printed Fixed-Wing Aircraft Competition
- Coordinating design goals and project objectives between subsystems to ensure seamless integration
- Translating competition requirements into design constraints to optimize aircraft weight, structural integrity, manufacturability, and flight performance

Tactical Autonomous Robotic Sentinel | *Sigma Phi Delta* Jan 2026 – Present

- Leading development of a proof-of-concept autonomous robot using servo-actuated joints
- Implementing voice-command recognition on an ESP32 to decode simple commands and map them to predetermined servo positions
- Designing a scalable mechatronics platform for future expansion into vision-based automation and artificial intelligence

Air Motor | *California Polytechnic State University* Nov 2025 – Dec 2025

- Collaborated on a five-person team to machine a functional compressed-air motor
- Manufactured precision components using five-axis CNC mills and manual lathes to meet dimensional tolerances within ± 0.005 in
- Assembled and tested the motor under controlled conditions, achieving speeds of approximately 2900 RPM

Secure Access Lockbox | *California Polytechnic State University* Nov 2024 – Dec 2024

- Collaborated on a three-person team to design an access-control system integrating an Arduino Mega, RFID authentication module, and keypad
- Utilized validation logic to authenticate credentials using RFID UID comparison and or pin validation
- Programmed LED validation behavior with an LCD status display and LED indicators

Vacuum Chamber | *Milwaukee Tool* Jan 2023 – Apr 2023

- Collaborated on a three-person team to design and prototype a 90 ft^3 vacuum chamber to enable in-house high-altitude testing
- Calculated pump sizing and chamber pressure requirements to reach 12 PSIA with an evacuation time of under 60 seconds.
- Developed proof-of-concept and detailed CAD models of the chamber structures, supported by strength and deflection analysis
- Achieved a vacuum capable of simulating altitudes up to 29,500 ft

Certifications

Certified SolidWorks Associate (CSWA) | *Dassault Systèmes*

Autodesk Inventor Certified User | *Certiport*

Lean Six Sigma Fundamentals | *LinkedIn*

Technical Skills

Software: SolidWorks, Autodesk Inventor, MATLAB, Siemens NX
Hardware: Hand Drafting, GD&T, Soldering, Machining (Manual Lathe & CNC Mill)